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Risk factors for COVID-19 infection, disease severity and related deaths in Africa: a systematic review

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Risk factors for COVID-19 infection, disease severity and related deaths in Africa: a systematic review

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ABSTRACT

Introduction: Coronavirus disease (COVID-19) has been identified for the first time in Wuhan Province, China in late December 2019. Understanding of the risk factors for infection and severe forms of the disease will contribute to contain the virus and its devastating consequences.

Methods: A systematic review has been conducted to synthesise existing evidence on risk factors affecting COVID-19 outcomes across Africa. We systematically searched MEDLINE, Scopus, MedRxiv and BioRxiv and performed a systematic narrative synthesis to describe the available studies for each outcome.

Results: Fifteen articles (15) met the inclusion criteria of which four were exclusively on Africa and the remaining 11 papers had a global focus with some data from Africa. Higher rates of infection in Africa are associated with high population density, urbanization, transport connectivity, high volume of tourism and international trade, and high level of economic and political openness. Limited or poor access to health care are also associated with higher COVID-19 infection rates. Older people and individuals with chronic conditions such as human immunodeficiency virus (HIV), tuberculosis and anaemia experience sever forms COVID-19 leading to hospitalisation and death. Similarly, high burden of chronic obstructive pulmonary disease (COPD), high prevalence of tobacco consumption, and low levels of expenditure on health and low levels of global health security score contribute to COVID-19 related deaths.

Conclusion: Demographic, institutional, ecologic, health system and politico-economic factors influenced the spectrum of COVID-19 infection, severity and death. We recommend multidisciplinary and integrated approaches to mitigate the identified factors and strengthen effective prevention strategies.

Key words: ABReN, Africa, COVID-19, death, infection, risk factors, severity, systematic review

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Strengths and limitations

- The present systematic review is focused on Africa and provided a comprehensive set of risk factors for the entire spectrum of COVID-19.
- The present findings underscore the need for a multidisciplinary and integrated approach to mitigate adverse effects of COVID-19 in Africa.
- There are limited studies of good quality on important risk factors, and inadequate data for meta-analysis.

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INTRODUCTION

The coronavirus disease (COVID-19) is a recently emerged viral disease caused by a novel single-stranded enveloped RNA (ribonucleic acid) virus, known as Severe Acute Respiratory Syndrome Coronavirus-2 (SARS-CoV-2)[1]. Since the first COVID-19 case was identified in Wuhan province in China, in late December 2019, the virus has rapidly spread across the world and declared a global pandemic on 11 March 2020[2]. As of 28 August 2020, the number of confirmed COVID-19 cases in the world has reached over 24 million with nearly 850 thousands deaths[3]. Despite the late start of the pandemic, infection rate in Africa has been increasing exponentially with a total of 1.5 million confirmed cases and about 30,00 deaths as of 28 August 2020[4]. The majority of reported cases in Africa were from South Africa and Egypt, while countries in Central Africa had recorded the least.

Given that the pandemic is caused by a novel strain of coronavirus with unknown original host[5], in the early stage of the outbreak the risk factors associated with its transmission routes, severity, and fatality risks were unclear. Several studies have been conducted to better understand the prognosis of the disease, though the level of uncertainty on viral shedding, transmissibility and disease severity remains high[6]. Evidence indicates that the susceptibility to infection, being seriously ill and the risk of death are influenced by individual-level characteristics such as socio-demographic factors[7], behavioural[8], and pre-existing medical conditions[9].

Whilst the number of people affected by the virus in Africa remains to be the lowest by global standards, which in part may be linked to the youthful age structure of the population and partly to the limited capacity for large scale testing[10], it is anticipated that the pandemic may have a profound impact in the region. First and foremost, Africa's health system is fragile. In addition, preventive strategies such as social distancing and hand washing, which have proved effective in reducing transmission in the rest of the world, are also likely to be less practical in the regions due to prevailing circumstances[11]. Compared to other regions, Africa is home to a large number of people living in sub-standard housing, which are mostly dense, inadequately ventilated and

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have limited access to direct water supply. The way of life in the region is largely communal. People tend to attend weddings, religious and funeral services in large numbers. Such societal and environmental factors along with the lack of awareness and access to preventive measures, and the potential for low level of adherence (even when the resources are available) mean that the region will have the highest number of vulnerable populations for COVID-19 infection [12].

Many countries have undertaken strict measures such as banning of public gatherings, complete lock-down of social and economic activities, and closure of borders to prevent importation of cases[13]. The number of new cases, however, continues to rise[11] creating a considerable pressure on the health-care system[14]. In the absence of effective vaccine or therapy, the ability to effectively control the spread of the pandemic, hence depends on effective monitoring of new cases, and on better understanding of the factors related to modes of transmission and severity of the disease. A comprehensive systematic review of African-focused COVID-19 related studies is required to strengthen the public health measures and response plan against the pandemic in the continent. This review aims to provide a comprehensive evidence on risk factors for transmission, disease severity, and COVID-19 related deaths in Africa.

METHODS

Australia Based Ethiopian Researchers Network (ABReN)

At the beginning of April, 2020, a time when the COVID-19 pandemic was spreading across the globe, a group of more than 50 Australian based Ethiopian researchers, who are active academic faculties in Australian institutions along with Ethiopian PHD students, established a working group on “Ethiopia and the COVID-19 pandemic” to examine various dimensions and consequences of COVID-19 in Ethiopia. Specifically, the working group focussed on four distinct thematic areas: (i) on understanding the epidemiology of COVID-19 in Ethiopia and around the globe and synthesising the lessons for the country, (ii) assessing opportunities for re-imagining the health system for COVID-19 pandemic response, (iii) identifying potential pharmaceutical and non-pharmaceutical interventions, and (iv) measuring socio-economic impact of COVID-19 in

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Ethiopia to support the Ethiopian government's effort to mitigate the deadly impact of the pandemic on the population. As part of the project under the first thematic area, we conducted a systematic review to synthesise the available evidence on COVID-19 in Africa on factors affecting infection rates, severity and related deaths.

Search strategy

This systematic review is reported according to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses guidelines [15]. Potentially eligible articles were systematically searched and retrieved from Medline, PubMed and Scopus up to 11 August 2020. The following search terms, tailored to each database were used: "COVID-19" OR "coronavirus disease" OR "SARS-COV-2" OR "2019 novel coronavirus" OR "2019-nCoV" AND "Africa". The full search strategy is summarised in Appendix 1. We used medical subject headings (MeSH) terms to identify synonyms and text words in the appropriate syntax of each database. The search was not limited by date of publication or study design. Reference lists of included studies were manually searched for additional relevant articles. Moreover, preprint articles from the MedRxiv and BioRxiv databases were also accessed to ensure wider coverage. However, search was restricted to studies conducted in Africa and the English language only.

Study selection criteria

The titles and abstracts of all publications identified by the search criteria were screened by two independent reviewers of the team (AA and BD) to retrieve full texts for all relevant studies. Full text articles were appraised using the Joanna Briggs Institute [16] critical appraisal tools[17] (Appendix 2). All studies published in English and reported at least one risk factor and/ or one health outcome were included in the review. We excluded letters to editors, correspondence, editorials, and case studies. However, we included global studies which reported findings from any country in Africa.

Socio-demographic, lifestyle and behavioural, climate variables and chronic comorbid conditions, which can be at a community or national level, were considered as risk factors. Whereas COVID-19 infection rates, level of severity, and COVID-19 related

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deaths were considered as outcomes. We adopted the definition of terms as reported in each of the included studies. Severity was referred for cases which required hospitalisation and are at high risk of death.

Data extraction

Seven members of the research team (HAG, DN, DF, MW, AG, BAD and AWA) extracted needed data using a standardised JBI data extraction form[17] (Appendix 3). The extracted data included study authors, setting, population characteristics, study design, outcome, and main findings on one or more risk factors and reported measure of association between risk factors and outcomes.

Quality and risk of bias assessment in individual studies

The JBI tools[17] were used to assess the methodological quality and risk of bias (in design, conduct and analysis) of included studies (Appendix 2). The tool is used to assess the inclusion criteria, measurement of exposure and outcome variables, confounding adjustment, and appropriateness of statistical analysis. All authors who participated in the data extraction also assessed the quality of studies that they extracted.

Data synthesis

A systematic narrative synthesis was conducted to describe the available studies separately for each outcome. Five steps were applied in synthesizing the data: (i) reading findings and results of each study systematically and comprehensively and assessing for relevance; (ii) developing a preliminary synthesis through describing each included study, grouping findings by infection, severity and death and tabulate results to identify patterns; (iii) organize factors linked to the outcomes described in phase (ii) and thematize items around themes; and (iv) explore the relationships of each factor within and among themes. Meta-analysis was not conducted because of the small number of studies identified for selected risk factors.

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RESULTS

Description of studies

The combined search terms identified 651 records (Figure 1). After removing duplicates (n=52) and unrelated studies (n=584), 53 records were eligible for full text screening. Finally, 15 studies [7-9 18-29] met the inclusion criteria and were included in this systematic review. Table 1 presents the main characteristics and outcomes of the included studies. Four studies were exclusively on Africa[9 18 21 22]. Two studies[7 29] assessed susceptibility to COVID-19 infection, nine studies[7 9 18 20 22-25 28] evaluated COVID-19 infection, three studies[7 19 21] examined severity of COVID-19 infection, and four studies[8 24 26 27] assessed COVID-19 related death. Four studies assessed more than one outcome[7 21 24 27]. As described in Table 1, most studies (13) were conducted among the general population. Study designs such as modelling (n=7), cross-sectional (n=2), genetics (n=1), survey (n=4) and Spatio-temporal analysis (n=1) were used in the included studies. The number of participants of those included studies for review ranges from 446 to 1.7 billion participants—the studies with the largest sample size were modelling related studies. We have annexed the quality score of all included studies in Appendix 4.

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Table 1. Characteristics of studies (n=15 studies) included in the systematic review

Author [30]	Setting	Population	Design	Sample (n)	COVID-19 Outcome	Exposure	Summary of study findings
Bergman et al (2020)[18]	Africa	Adults	Modelling	446	Infection	Mobility	The authors reported that there was no statistically significant association between transmission rates and lagged mobility. The authors also added that the information is limited to Google workplace mobility as Apple mobility data is limited in Africa unlike other regions
Clark et al (2020)[19]	Africa/Global	General	Modelling	1.7 Billion	Severity	Age, sex, comorbidity and continent	Individuals who are diagnosed with at least one underlying condition are at risk of severe COVID-19 disease. A total of 1.17 billion (22%) of the global population were estimated to be at risk of sever COVID-19 disease, and the prevalence of one or more co-moribid condition was approximately 10% by age 25 years, 33% by 50 years, and 66% by 70 years. There was no difference by sex. Individuals who are diagnosed with multi-morbidities are at a higher risk of severe COVID-19 disease. A total of 400 million (6%) of the global population were estimated to be at risk of sever COVID-19 disease, and is higher in older people. The most prevalent conditions in those aged 50+ years were CKD, CVD, CRD and diabetes. The prevalence of multimorbidity (2+ underlying chronic illness) was three times higher in Europe than in Africa (10% vs 3%).
Daon et al(2020)[20]	Africa/Global	General	Modelling	1364 major airports (connections)	Infection	City connectivity	An outbreak of COVID-19 in Africa is most likely to originate from a passenger who travelled from Europe, and the average risk of COVID-19 infection originated from airport in Africa is 0.06, lower result compared to Airports in Asia with an estimated risk of 0.51). The airport with the highest risk to initiate an outbreak in Africa is in Johannesburg with a risk estimate of 0.1.

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Author [30]	Setting	Population	Design	Sample (n)	COVID-19 Outcome	Exposure	Summary of study findings
Davies et al (2020)[7]	Africa/Global	General	Modelling	-	Infection, Severity	Age	Children are at a lower rate of infection and severity compared to older individuals. The study revealed that people younger than 20 years old are roughly half susceptible as those who are older than 20 years old. In addition, 75% of infections are subclinical in children ageing between 10 and19 years as compared to 24% in people aged 70 years and above.
Diop et al (2020)[21]	Africa	General	Modelling	-	Infection Severity	Residence, population density, comorbidity (HIV, Tb and Anemia)	The rate of COVID-19 infection increases as the population density increases. Linked to this, the rate of infection is low in rural settings than in urban areas. The study reported that comorbidity would increase severity of the disease. The rate of anemia, TB & HIV compared to the rate of DM, HTN and obesity is higher in Africa than other continents, and these (anemia, TB and HIV) are the underlying conditions which lead to COVID-19 related severity in Africa.
Gayawan et al (2020)[22]	Africa	General	Spatio-temporal analysis	Not specified	Infection	number of physicians number of hospital beds	Positive correlation was revealed between COVID-19 and number of physicians (r=0.49, p-value=<0.001) and hospital beds (r=0.14, p-value=0.34) though only the estimate for physicians is significant. The Spatio-temporal analysis reveals that the occurrence and burden of COVID-19 in Africa varied geographically with neighboring countries particularly in the western part of the continent which could imply that neighboring countries pose significant importation risk to each other
Hossain et al (2020)[23]	Africa/Global	General	Modelling	1,174,652	Infection	Weather, economic openness, and political democracy	The study revealed that countries with warmer temperatures showed slower spread than their comparator. Countries with higher economic openness had higher infection rates than with lower economic openness. The countries with stronger

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Author [30]	Setting	Population	Design	Sample (n)	COVID-19 Outcome	Exposure	Summary of study findings
							political democracy had higher infection rates than with weaker political democracy.
Jaffé et al (2020)[24]	Africa/Global	General	Cross-sectional	Dataset of 36 countries	Infection Mortality	imports of goods and services, international tourism, electric power consumption, and Population over 65 years old	The study revealed that globalized countries are at risk of infection and, globalized low-income countries are at higher risk of mortality. Countries with higher imports of goods and services, and international tourism had higher infection rates than their comparator
Kubota *et al* (2020)[25]	Africa/Global	General	Modelling study	1,055 countries	Infection	Age, precipitation, temperature, international travel, BCG vaccination, malaria infection	The study found a negative correlation between the accumulated numbers of the COVID-19 cases and the following variables: mean temperature, BCG vaccination effect and malaria infection. The study found a positive correlation between the accumulated numbers of the COVID-19 cases and the following variables: mean precipitation, the relative frequency of foreign visitors per population, GDP per person, population density, and relative proportion of people ≥ 65 years old.
Maitra et al (2020)[26]	Africa/Global	General	Cross sectional	422,582	Mortality	Health expenditure	The study found that countries with low health expenditure (percentage of GDP) were significantly associated with higher case fatality rate (p=0.0017).
Maraghi et al. (2020)[27]	Africa/Global	Not specified	Cross-sectional	>1000 but globally	Case fatality, recovery rate	Prevention, detection, response, risk environment indices and global health security (GHS)	Prevention (r=-0.988, -0.1.0 to -0.548), p= 0.012), detection (r=-0.979 (-1.000, -0.312), p=0.021), response (r=-0.965 (-0.999, -0.051), p=0.035), and GHS (r=-0.995 (-1.000, -0.786), P = 0.005) were negatively correlated with case fatality rate (CFR) in African countries.
Muneer *et al* (2020)[28]	Africa/Global	General	Web-based survey (worldometer)	-	Infection	Malaria prevalence	The study demonstrated that a high rate of malaria was negatively correlated with Covid-19 (r= -0.15, p= 0.02).
Okpokoro	Africa/Global	General	Ecologic study		Mortality	COPD, Tobacco use	The study revealed that mortality was positively correlated with COPD (Rho=-0.28, p=0.09) and

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Author [30]	Setting	Population	Design	Sample (n)	COVID-19 Outcome	Exposure	Summary of study findings
et al (2020)[8]							Tobacco use (Rho=-0.01, p=0.91) although not statistically significant. Similarly, the study showed that mortality was negatively correlated with life expectancy (Rho=-0.09, p=0.24), quality of air (Rho=-0.02, p=0.84), and life expectancy (Rho=-0.24, p=0.1).
Onovo et al (2020)[9]	Sub-Saharan Africa (SSA)	General	Survey	-	Infection	Adult HIV incidence rate, infant's pneumococcal conjugate-based vaccine, incidence of malaria, diarrhea treatment for under 5	The study found that HIV incidence (p= 0.0001), pneumococcal conjugate-based vaccine (p= 0.001), incidence of malaria (p= 0.001) and diarrhea treatment (p= 0.002) could significantly predict COVID-19 infection in sub-Saharan Africa (SSA). However, TB incidence (p= 0.86) and smoking prevalence (p= 0.53) were not statistically associated with coronavirus disease infection.
Ortiz-Fernández et al (2020)[29]	Africa/Global	General	Genetics	2,504 .	Susceptibility	Lower expression of angiotensin-converting enzyme 2 (ACE2) and type II transmembrane serine protease (TMPRSS2) genes	The study revealed that Africans have lower expression levels of ACE2 and TMPRSS2, suggesting a lower susceptibility for SARS-CoV-2 infection in the African populations.

CKD: chronic kidney disease; COPD: chronic obstructive pulmonary disease; CRD: chronic respiratory disease; CVD: cardiovascular diseases; DM= diabetes mellitus; GDP: gross domestic products; HTN= hypertension

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Factors affecting COVID-19 infection, severity, and related death

Demographic, institutional, politico-economic, environmental, health system policy, lifestyle factors and presence and level of chronic conditions were reported to have an effect on prevention and spread of COVID-19[7-9 18-29]. For example, we found that COVID-19 infection and related deaths were affected by politico-economic factors such as economic openness and political democracy, a high volume of imports of goods and services and international tourism. Furthermore, high rate of comorbid conditions was positively correlated with COVID-19 infection and COVID-19 related death, with people having one or more chronic conditions experiencing severe form of the disease. A detailed summary of all the relevant factors affecting the infection, severity, and mortality related to COVID-19 is presented in Table 2.

Table 2. Factors associated with COVID-19 infection, severity and related death in Africa

Outcome	Theme	Potential specific factors
Infection	Demographic factors	High number of people aged ≥65 years[25], high population density[21 25]
	Health system level factors	Low number of physicians and hospitals[22], low BCG vaccination[24]
	Politico-economic factors	High economic openness, strong political democracy[23], more imports of goods and services, wide international tourism, high international travel[24], and city connectivity[20],
	Environmental factor	Low mean temperature[23 25]
	Comorbid conditions	Low malaria infection rate[9 25 28], high HIV infection rate and under-5 diarrheal disease[9]
Severity	Demographic factors	High percentage of older age population profile[7 19]
	Comorbid conditions	People with HIV, TB and anemia [19 21]
	Health system policy factors	Low prevention, detection, response and risk environment indices[27]
Death	Politico-economic factors	More imports of goods and services, wide international tourism[24], low global health security index score[27] and low health expenditure (percentage of GDP)[26]
	Chronic illness and lifestyle factors	High prevalence of chronic illness and Tobacco use[8]

BCG: Bacillus Calmette–Guérin; HIV: Human Immunodeficiency Virus; TB: Tuberculosis; GDP: Gross Domestic Product; COPD: Chronic Obstructive Pulmonary Disease

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DISCUSSION

The systematic review presents a comprehensive summary of the current evidence on risk factors of COVID-19 infection, severity, and related deaths in Africa. Demographic, institutional, political, and ecological factors were linked with high COVID-19 infection rates in Africa. The study also revealed that severe forms of COVID-19 were associated with comorbidities and specific demographic characteristics. Health system organization and policy, politico-economic situation, prevalence of chronic conditions, and lifestyle factors increase the risk of deaths associated with COVID-19.

African countries which heavily rely on imported goods and services and international tourism were likely to have higher infection rates than other countries in the region [23] [24]. Unsurprisingly, the rate of COVID-19 infection was higher among densely populated countries. Densely populated are conducive for transmission give that the disease is mainly spread through close contact with infected persons [3132].

African countries enjoying a stable political democracy, have an open economy and well connected air and road network were found to have a higher risk of COVID-19 infection [20] [23]. This could be attributed to the fact that countries with strong democracy have a laissez-faire administrative system, where people are not obliged to implement some public health measures which infringe individual freedom. Democratic systems also go hand in hand with law and order and a higher than average disposable income, which create opportunities for leisure and work-related travels both internally and internationally, which increase the risk of COVID-19 transmission. In addition, countries with open economies and high city connectivity are characterised by high mobility, facilitating spread of the disease.

The studies included in this review reported mixed findings with regards to the effect of temperature on COVID-19 infection. For example, the current review indicated that people living in countries with warmer temperature are less likely to acquire infection [23] [33]. Contrary to this, African countries with warm weather such as Egypt are highly affected by the virus as compared to other African countries with relatively cooler temperature[34]. In contrast, Yao and colleagues [35] demonstrated that temperature had no significant association with COVID-19 infection. Further investigation is required to determine the role of weather on COVID-19 spread.

Being an older age was associated with more severe forms of COVID-19 at admission to hospital[36]. This might be explained by weaker immunity among older people. Patients with

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co-infections such as TB are more likely to experience severe forms of COVID-19 [21]. TB infection is common in low-resource settings and among older people with pre-existing conditions resulting in high vulnerability to severe form of COVID-19 infection[19 37]. Evidence on the effect of COVID-19 on people with HIV and malaria is limited [38]. A study by Karmen et al showed that being infected with HIV at the time of acquiring COVID-19 does not significantly increase the severity of illness or the risk of complications [39]. However, further research is warranted as the available evidence is inconclusive.

Generally, patients with underlining chronic conditions are at a higher risk of having sever COVID-19 and related mortality. For instance, a study conducted by Cox et al[38] revealed greater risk and severe forms of COVID-19 in patients with COPD. Findings of a meta-analysis of seven studies conducted in China also reported significant association between COVID-19 and hypertension, chronic respiratory disease, and cardiovascular disease [40]. These findings imply that countries with a high burden of chronic conditions are more likely to have more severe cases of COVID-19. People who died of COVID-19 had chronic hypertension, COPD and cardiovascular comorbidities than recovered patients [41 8]. History of tobacco use was found to be associated with increased risk of COVID-19 death [8 42]. Additionally, Zhou and colleagues [43] found that 50% of patients who died of COVID-19 had a history of secondary bacterial infections. Chen and colleagues [44] also reported that bacterial and fungal co-infections increase the risk of COVID-19 related mortality. Thus, public health and medical services responding to the pandemic in Africa should be equipped with resources that enable the identification of cases with underlining chronic conditions and provide tailored interventions.

The present review revealed that countries with effective health care system such as a strong disease prevention, case detection and response program and have a strong global health security system in pace had lower case fatality rate[27]. This was supported by a WHO report that the COVID-19 pandemic is straining health systems worldwide as health facilities have been overstretched and unable to operate effectively due to increased burden for health facilities and health care workers [45]. Importantly, improved health expenditure significantly reduces the risk of COVID-19 deaths [26]. This could be attributed to the fact that a strong health system with sufficient resources is both adaptive, resilient and often has the required manpower and facilities to respond to the higher demand for medical staff and supplies in treating COVID-19 cases.

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The study has the following limitations. First, the results reported in the paper with respect to associated factors do not necessarily imply causality as most of the included studies were based on either cross-country analyses or cross-sectional designs that are not suitable for studying causal relationships. Second, included articles were limited to those written in English. Third, there is still limited evidence on COVID-19 in countries Africa (some of the included studies are even pre-print studies) and the review was unable to provide a stratified analysis by regions.

CONCLUSIONS

This systematic review demonstrated that several demographic, institutional, politico-economic, environmental factors and comorbid conditions increased the risk of COVID-19 infection in Africa. Severe forms of COVId-19 and deaths related to COVID-19 were linked to health system resources, politico-economic circumstances, the presence of comorbid conditions, and lifestyle factors. These aspects should be factored in by African Governments and their development partners while designing tailored and targeted interventions to mitigate effects of the disease in the region. Implementing lock-down measures in African countries is difficult or at least not as easy as in western countries, due to the region's unique social and economic settings. Alternative and effective measures such as mandatory face mask use and strong contact tracing system could be the most feasible interventions to contain the pandemic in the region [46].

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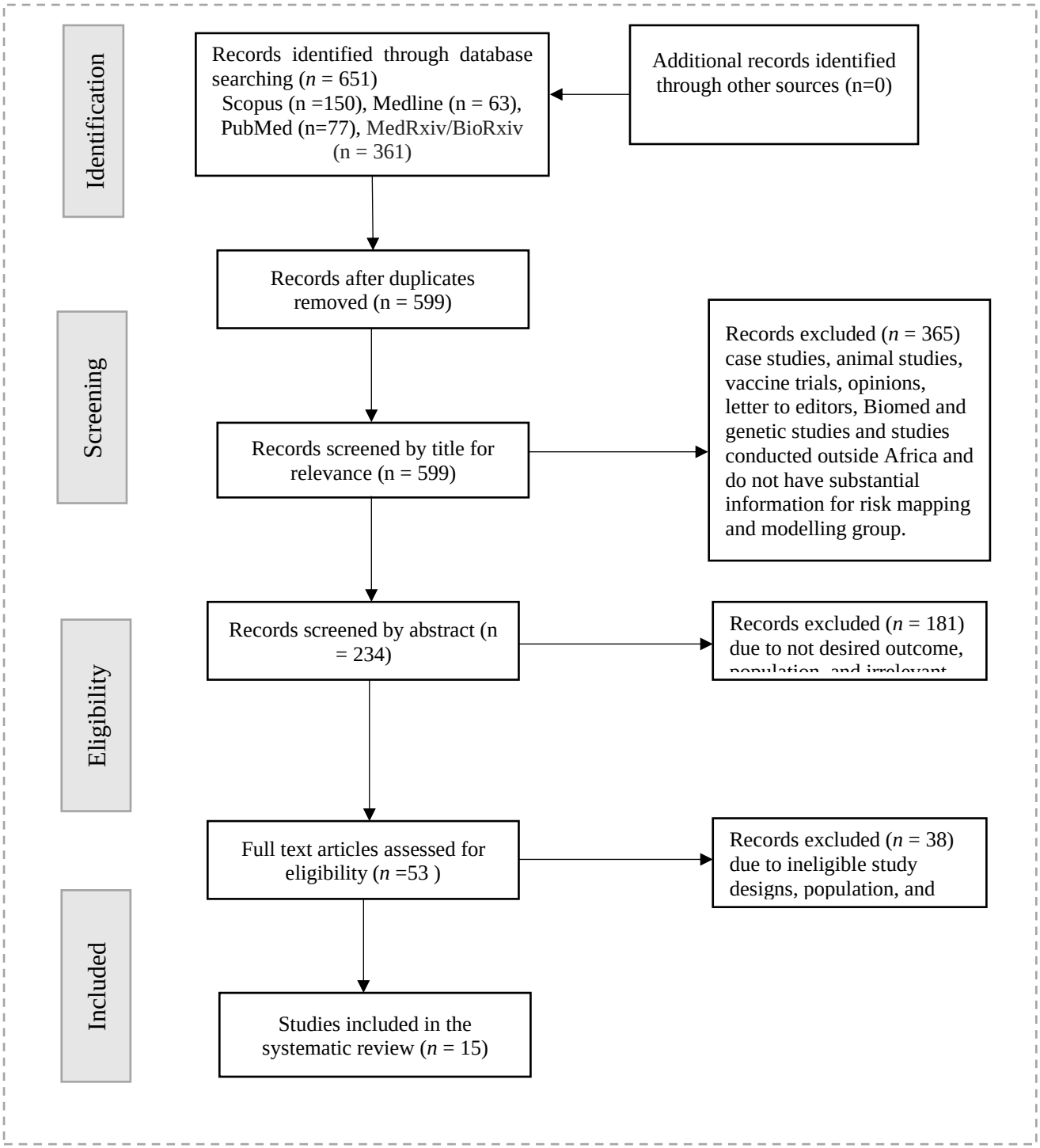


Figure 1. PRISMA Flow Diagram

Appendix 1: Search strategy

No	Terms/keywords
Medline	
1	Coronavirus/ or Coronavirus Infections/ or COVID-19.mp
2	coronavirus.mp
3	SARS-CoV-2.mp
4	n-CoV.mp
5	2019-nCoV.mp
6	1 or 2 or 3 or 4 or 5
7	Africa.mp. or exp Africa/
8	6 and 7
9	limit 15 to (English language and humans and year="2019 - 2020")
PubMed	
1	"COVID-19" OR coronavirus OR "SARS-CoV-2" OR "n-CoV" OR "2019-nCoV"
2	Africa*
3	(("COVID-19" OR coronavirus OR "SARS-CoV-2" OR "n-CoV" OR "2019-nCoV")) AND (Africa)
4	(("COVID-19" OR coronavirus OR "SARS-CoV-2" OR "n-CoV" OR "2019-nCoV")) AND (Africa) Sort by: Best Match Filters: Publication date from 2019/12/01; Humans; English
Scopus	(TITLE-ABS-KEY ("COVID-19" OR coronavirus OR "SARS-CoV-2" OR "n-CoV" OR "2019-nCoV")) AND (TITLE-ABS-KEY (africa) AND (LIMIT-TO (PUBYEAR , 2020) OR LIMIT-TO (PUBYEAR , 2019))

*list of countries was considered

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Appendix 2: JBI quality appraisal and selection tool

JBI Critical Appraisal Checklist for Descriptive / Case Series

Reviewer _____ Date _____

Author _____ Year _____ Record Number _____

	Yes	No	Unclear	Not Applicable
1. Was study based on a random or pseudo-random sample?	☐	☐	☐	☐
2. Were the criteria for inclusion in the sample clearly defined?	☐	☐	☐	☐
3. Were confounding factors identified and strategies to deal with them stated?	☐	☐	☐	☐
4. Were outcomes assessed using objective criteria?	☐	☐	☐	☐
5. If comparisons are being made, was there sufficient descriptions of the groups?	☐	☐	☐	☐
6. Was follow up carried out over a sufficient time period?	☐	☐	☐	☐
7. Were the outcomes of people who withdrew described and included in the analysis?	☐	☐	☐	☐
8. Were outcomes measured in a reliable way?	☐	☐	☐	☐
9. Was appropriate statistical analysis used?	☐	☐	☐	☐

Overall appraisal: Include ☐ Exclude ☐ Seek further info ☐

Comments (Including reason for exclusion)

Appendix 3: JBI data extraction tool

JBI Data Extraction Form for Experimental / Observational Studies

Reviewer _____ Date _____

Author _____ Year _____

Journal _____ Record Number _____

Study Method

RCT ☐ Quasi-RCT ☐ Longitudinal ☐

Retrospective ☐ Observational ☐ Other ☐

Participants

Setting _____

Population _____

Sample size

Group A _____ Group B _____

Interventions

Intervention A _____

_____

Intervention B _____

_____

Authors Conclusions:

_____

_____

_____

Reviewers Conclusions:

_____

_____

_____

Appendix 4: Assessment of methodological quality (n=15)

Authors, year	***Cross-sectional studies***									
	Was study based on a random or pseudo-random sample?	Were the criteria for inclusion in the sample clearly defined?	Were confounding factors identified and strategies to deal with them stated?	Were outcomes assessed using objective criteria?	If comparisons are being made, was there sufficient comparison of groups?	Was the follow-up carried out over a sufficient period?	Were the outcomes of people who withdrew described and included in the analysis?	Were outcomes measured in reliable way?	Was appropriate statistical analysis used?	%
Bergman et al (2020)[18]	N	NA	Y	Y	NA	Y	NA	U	Y	80
Clark et al (2020)[19]	NA	Y	Y	Y	NA	NA	NA	Y	Y	100
Daon et al(2020)[20]	NA	Y	Y	Y	NA	NA	NA	Y	Y	100
Davies et al (2020)[7]	NA	Y	Y	Y	NA	NA	NA	Y	Y	100
Diop et al (2020)[21]	NA	Y	Y	Y	NA	NA	NA	Y	Y	100
Hossain et al (2020)[23]	N	Y	U	Y	Y	N	NA	Y	Y	72
Jaffé et al (2020)[24]	N	Y	U	Y	Y	N	NA	Y	Y	72
Kubota *et al* (2020)[25]	N	Y	Y	Y	Y	N	NA	Y	Y	86
Maitra et al (2020)[26]	N	Y	Y	Y	NA	NA	NA	Y	Y	83
Gayawan et al (2020)[22]	N	Y	N	Y	Y	N/A	N/A	Y	Y	80%
Maraghi et al. (2020)[27]	Yes	Yes	Yes	No	Yes	NA	NA	No	Yes	85.7
Muneer *et al* (2020)[28]	Y	Y	N	Y	Y	NA	NA	Y	Y	86
Okpokoro et al (2020)[8]	Y	Y	N	Y	Y	NA	NA	Y	Y	86
Onovo et al (2020)[9]	Y	Y	Y	Y	Y	NA	NA	Y	Y	100
Ortiz-Fernández et al (2020)[29]	Y	Y	N	Y	N	NA	NA	Y	Y	72

Y=Yes; N= No; U= Unclear; NA= Not applicable; %= Percentage of score

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Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	3
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	5-6
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	6
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	No protocol
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	7
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	7
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	7
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	7
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	8
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	7-8
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	8
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	8
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	8

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Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	8
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	Not Applicable
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	9
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	9
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	9, and Appendix 4
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	10-13
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	Not Applicable
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	Not Applicable
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	15-16
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	17
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	17
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	17

From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(6): e1000097. doi:10.1371/journal.pmed1000097

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Risk factors for COVID-19 infection, disease severity and related deaths in Africa: a systematic review

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Primary Subject Heading:	Epidemiology
Secondary Subject Heading:	Infectious diseases
Keywords:	Public health < INFECTIOUS DISEASES, EPIDEMIOLOGY, RESPIRATORY MEDICINE (see Thoracic Medicine)

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Risk factors for COVID-19 infection, disease severity and related deaths in Africa: a systematic review

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ABSTRACT

Objective: The aim of this study was to provide a comprehensive evidence on risk factors for transmission, disease severity, and COVID-19 related deaths in Africa.

Design: A systematic review has been conducted to synthesise existing evidence on risk factors affecting COVID-19 outcomes across Africa.

Data sources: Data were systematically searched from MEDLINE, Scopus, MedRxiv and BioRxiv.

Eligibility criteria: Studies for review were included if they were published in English and reported at least one risk factor and/ or one health outcome. We included all relevant literature published up until 11 August 2020.

Data extraction and synthesis: We performed a systematic narrative synthesis to describe the available studies for each outcome. Data were extracted using a standardised JBI data extraction form.

Results: Fifteen articles met the inclusion criteria of which four were exclusively on Africa and the remaining 11 papers had a global focus with some data from Africa. Higher rates of infection in Africa are associated with high population density, urbanization, transport connectivity, high volume of tourism and international trade, and high level of economic and political openness. Limited or poor access to health care are also associated with higher COVID-19 infection rates. Older people and individuals with chronic conditions such as human immunodeficiency virus (HIV), tuberculosis and anaemia experience severe forms COVID-19 leading to hospitalisation and death. Similarly, high burden of chronic obstructive pulmonary disease (COPD), high prevalence of tobacco consumption, and low levels of expenditure on health and low levels of global health security score contribute to COVID-19 related deaths.

Conclusions: Demographic, institutional, ecologic, health system and politico-economic factors influenced the spectrum of COVID-19 infection, severity and death. We recommend multidisciplinary and integrated approaches to mitigate the identified factors and strengthen effective prevention strategies.

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Key words: ABReN, Africa, COVID-19, death, infection, risk factors, severity, systematic review

Strengths and limitations

- Effective public health interventions against the COVID-19 pandemic require strong evidence on the risk factors associated with the disease.
- The present study is the first African-focused comprehensive systematic review of its kind on a wide range of risk factors associated with the entire spectrum of COVID-19 in the Africa region.
- Titles and abstracts of all publications identified by the search criteria were screened by two independent reviewers of the team.
- Full text articles were appraised using the Joanna Briggs Institute critical appraisal tools.
- The available data did not allow us to carry out meta-analysis nor firmly establish causal relationships in the present study.

INTRODUCTION

The coronavirus disease (COVID-19) is a recently emerged viral disease caused by a novel single-stranded enveloped RNA (ribonucleic acid) virus, known as Severe Acute Respiratory Syndrome Coronavirus-2 (SARS-CoV-2)[1]. Since the first COVID-19 case was identified in Wuhan province in China, in late December 2019, the virus has rapidly spread across the world and declared a global pandemic on 11 March 2020[2]. As of 21 November 2020, the number of confirmed COVID-19 cases in the world has reached over 55.6 million with nearly 1.34 million deaths[3]. Despite the late start of the pandemic, infection rate in Africa has been increasing exponentially with a total of 2.01 million confirmed cases and more than 40 thousand deaths as of 21 November 2020[4]. The majority of reported cases in Africa were from South Africa and Egypt, while countries in Central Africa had recorded the least.

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Given that the pandemic is caused by a novel strain of coronavirus with unknown original host[5], in the early stage of the outbreak the risk factors associated with its transmission routes, severity, and fatality risks were unclear. Several studies have been conducted to better understand the prognosis of the disease, though the level of uncertainty on viral shedding, transmissibility and disease severity remains high[6]. Evidence indicates that the susceptibility to infection, being seriously ill and the risk of death are influenced by individual-level characteristics such as socio-demographic factors[7], behavioural traits[8], and pre-existing medical conditions[9].

Whilst the number of people affected by the virus in Africa remains to be the lowest by global standards, which in part may be linked to the youthful age structure of the population and partly to the limited capacity for large scale testing[10], it is anticipated that the pandemic may have a profound impact in the region. First and foremost, Africa's health system is fragile. In addition, preventive strategies such as social distancing and hand washing, which have proved effective in reducing transmission in the rest of the world, are less likely to be practical in the region due to prevailing socio-cultural and economic circumstances[11]. Compared to other regions, Africa is home to a large number of people living in sub-standard housing, which are mostly dense, inadequately ventilated and have limited access to direct water supply. The way of life in the region is largely communal. People tend to attend weddings, religious and funeral services in large numbers. Such societal and environmental factors along with the lack of awareness and access to preventive measures, and the potential for low level of adherence (even when the resources are available) mean that the region will have the highest number of vulnerable populations for COVID-19 infection [12].

Many countries have undertaken strict measures such as banning of public gatherings, complete lock-down of social and economic activities, and closure of borders to prevent importation of cases[13]. The number of new cases, however, continues to rise[11] creating considerable pressure on the health-care system[14]. In the absence of effective vaccine or therapy, the ability to effectively control the spread of the pandemic, hence depends on effective monitoring of new cases, and on better understanding of the factors

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related to modes of transmission and severity of the disease. A comprehensive systematic review of African-focused COVID-19 related studies is required to strengthen public health measures and response plan against the pandemic in the continent. This review aims to provide a comprehensive evidence on risk factors for transmission, disease severity, and COVID-19 related deaths in Africa.

METHODS

Australia Based Ethiopian Researchers Network (ABReN)

At the beginning of April, 2020, a time when the COVID-19 pandemic was spreading across the globe, a group of more than 50 Australian based Ethiopian researchers, who are active academic faculties in Australian institutions along with Ethiopian PHD students, established a working group on "Ethiopia and the COVID-19 pandemic" to examine various dimensions and consequences of COVID-19 in Ethiopia. Specifically, the working group focussed on four distinct thematic areas: (i) on understanding the epidemiology of COVID-19 in Ethiopia and around the globe and synthesising the lessons for the country, (ii) assessing opportunities for re-imagining the health system for COVID-19 pandemic response, (iii) identifying potential pharmaceutical and non-pharmaceutical interventions, and (iv) measuring socio-economic impact of COVID-19 in Ethiopia to support the Ethiopian government's effort to mitigate the deadly impact of the pandemic on the population. As part of the project under the first thematic area, we conducted a systematic review to synthesise the available evidence on COVID-19 in Africa on factors affecting infection rates, severity, and related deaths.

Search strategy

This systematic review is reported according to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses guidelines [15]. Potentially eligible articles were systematically searched and retrieved from Medline, PubMed and Scopus up to 11 August 2020. The following search terms, tailored to each database were used: "COVID-19" OR "coronavirus disease" OR "SARS-COV-2" OR "2019 novel coronavirus" OR "2019-nCoV" AND "Africa". The full search strategy is summarised in Appendix 1. We used medical

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subject headings (MeSH) terms to identify synonyms and text words in the appropriate syntax of each database. The search was not limited by date of publication or study design. Reference lists of included studies were manually searched for additional relevant articles. Moreover, preprint articles from the MedRxiv and BioRxiv databases were also accessed to ensure wider coverage. However, search was restricted to studies conducted in Africa and the English language only.

Study selection criteria

The titles and abstracts of all publications identified by the search criteria were screened by two independent reviewers of the team (AAA and BAD) to retrieve full texts for all relevant studies. Full text articles were appraised using the Joanna Briggs Institute [16] critical appraisal tools[17] (Appendix 2). All studies published in English and reported at least one risk factor and/ or one health outcome were included in the review. We excluded letters to editors, correspondence, editorials, and case studies. However, we included global studies which reported findings from any country in Africa.

Socio-demographic, lifestyle and behavioural, climate variables and chronic comorbid conditions, which can be at a community or national level, were considered as risk factors. Whereas COVID-19 infection rates, level of severity, and COVID-19 related deaths were considered as outcomes. We adopted the definition of terms as reported in each of the included studies. Severity was referred for cases which required hospitalisation and are at high risk of death.

Data extraction

Seven members of the research team (HAG, DNK, DMF, MW, ATG, BAD and AAA) extracted needed data using a standardised JBI data extraction form[17] (Appendix 3). The extracted data included study authors, setting, population characteristics, study design, outcome, and main findings on one or more risk factors and reported measure of association between risk factors and outcomes.

Quality and risk of bias assessment in individual studies

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The JBI tools[17] were used to assess the methodological quality and risk of bias (in design, conduct and analysis) of included studies (Appendix 2). The tool is used to assess the inclusion criteria, measurement of exposure and outcome variables, confounding adjustment, and appropriateness of statistical analysis. All authors who participated in the data extraction also assessed the quality of studies that they extracted.

Data synthesis

A systematic narrative synthesis was conducted to describe the available studies separately for each outcome. Five steps were applied in synthesizing the data: (i) reading findings and results of each study systematically and comprehensively and assessing for relevance; (ii) developing a preliminary synthesis through describing each included study, grouping findings by infection, severity and death and tabulate results to identify patterns; (iii) organize factors linked to the outcomes described in phase (ii) and thematize items around themes; and (iv) explore the relationships of each factor within and among themes. Meta-analysis was not conducted because of the small number of studies identified for selected risk factors.

Patient and public involvement

Patients and/or the public were not involved in the design, or conduct, or reporting, or dissemination plans of this research.

RESULTS

Description of studies

The combined search terms identified 651 records (Figure 1). After removing duplicates (n=52) and unrelated studies (n=584), 53 records were eligible for full text screening. Finally, 15 studies [7-9 18-29] met the inclusion criteria and were included in this systematic review. Table 1 presents the main characteristics and outcomes of these studies. Four studies were exclusively on Africa[9 18 21 22]. Two studies[7 29] assessed susceptibility to COVID-19 infection, nine studies[7 9 18 20 22-25 28] evaluated COVID-19 infection, three studies[7 19 21] examined severity of COVID-19 infection, and four studies[8 24 26 27] assessed COVID-19 related death. Four studies assessed more than one outcome[7 21 24 27]. As described in Table 1, most studies (13) were conducted among the general population. Study designs

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such as modelling (n=7), cross-sectional (n=2), genetics (n=1), survey (n=4) and Spatio-temporal analysis (n=1) were used in the included studies. The number of participants of those included studies for review ranges from 446 to 1.7 billion participants—the studies with the largest sample size were modelling related studies. We have annexed the quality score of all included studies in Appendix 4.

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Table 1. Characteristics of studies (n=15 studies) included in the systematic review

Author [30]	Setting	Population	Design	Sample (n)	COVID-19 Outcome	Exposure	Summary of study findings
Bergman et al (2020)[18]	Africa	Adults	Modelling	446	Infection	Mobility	There was no statistically significant association between transmission rates and lagged mobility. But there is limited information about Google workplace mobility data and COVID-19 infection as Apple mobility data is limited in Africa unlike other regions.
Clark et al (2020)[19]	Africa/Global	General	Modelling	1.7 Billion	Severity	Age, sex, comorbidity and continent	Individuals who are diagnosed with at least one underlying condition are at risk of severe COVID-19 disease. Although not different by sex, a total of 1.17 billion (22%) of the global population were estimated to be at risk of sever COVID-19 disease, and the prevalence of one or more co-moribid condition was approximately 10% by age 25 years, 33% by 50 years, and 66% by 70 years. A total of 400 million (6%) of the global population were estimated to be at risk of sever COVID-19 disease, and is higher in older people. The most prevalent conditions in those aged 50+ years were CKD, CVD, CRD and diabetes— the prevalence of multimorbidity (2+ underlying chronic illness) was three times higher in Europe than in Africa (10% vs 3%).
Daon et al(2020)[20]	Africa/Global	General	Modelling	1364 major airports (connections)	Infection	City connectivity	An outbreak of COVID-19 in Africa is most likely to originate from a passenger who travelled from Europe, and the average risk of COVID-19 infection originated from airport in Africa is 0.06, lower result compared to Airports in Asia with an estimated risk of 0.51). The airport with the highest risk to initiate an outbreak in Africa is in Johannesburg with a risk estimate of 0.1.
Davies et al (2020)[7]	Africa/Global	General	Modelling	-	Infection, Severity	Age	People younger than 20 years old are roughly half susceptible as those who are older than 20 years old. In addition, 75% of infections are subclinical in

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Author [30]	Setting	Population	Design	Sample (n)	COVID-19 Outcome	Exposure	Summary of study findings
							children ageing between 10 and19 years as compared to 24% in people aged 70 years and above.
Diop et al (2020)[21]	Africa	General	Modelling	-	Infection Severity	Residence, population density, comorbidity (HIV, Tb and Anemia)	The rate of COVID-19 infection increases as the population density increases. Linked to this, the rate of infection is low in rural settings than in urban areas. The study reported that comorbidity would increase severity of the disease. The rate of anemia, TB & HIV compared to the rate of DM, HTN and obesity is higher in Africa than other continents, and these (anemia, TB and HIV) are the underlying conditions which lead to COVID-19 related severity in Africa.
Gayawan et al (2020)[22]	Africa	General	Spatio-temporal analysis	Not specified	Infection	number of physicians number of hospital beds	Positive correlation was revealed between COVID-19 and number of physicians (r=0.49, p-value=<0.001) and hospital beds (r=0.14, p-value=0.34) though only the estimate for physicians is significant. The Spatio-temporal analysis reveals that the occurrence and burden of COVID-19 in Africa varied geographically with neighboring countries particularly in the western part of the continent which could imply that neighboring countries pose significant importation risk to each other
Hossain et al (2020)[23]	Africa/Global	General	Modelling	1,174,652	Infection	Weather, economic openness, and political democracy	Countries with warmer temperatures showed slower spread than their comparator. Countries with higher economic openness had higher infection rates than with lower economic openness. The countries with stronger political democracy had higher infection rates than with weaker political democracy.
Jaffé et al (2020)[24]	Africa/Global	General	Cross-sectional	Dataset of 36 countries	Infection Mortality	imports of goods and services, international tourism, electric power consumption,	Countries that were connected to the global economy tended to have a higher risk of infection, while the risk of mortality from the disease are higher in less globalized low-income countries. Countries with higher imports of goods and services, and

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Author [30]	Setting	Population	Design	Sample (n)	COVID-19 Outcome	Exposure	Summary of study findings
						and Population over 65 years old	international tourism had higher infection rates than their comparator.
Kubota *et al* (2020)[25]	Africa/Global	General	Modelling study	1,055 countries	Infection	Age, precipitation, temperature, international travel, BCG vaccination, malaria infection	A negative correlation between the accumulated numbers of the COVID-19 cases and the following variables: mean temperature, BCG vaccination effect and malaria infection. The study found a positive correlation between the accumulated numbers of the COVID-19 cases and the following variables: mean precipitation, the relative frequency of foreign visitors per population, GDP per person, population density, and relative proportion of people ≥ 65 years old.
Maitra et al (2020)[26]	Africa/Global	General	Cross sectional	422,582	Mortality	Health expenditure	Countries with low health expenditure (percentage of GDP) were significantly associated with higher case fatality rate (p=0.0017).
Maraghi et al. (2020)[27]	Africa/Global	Not specified	Cross-sectional	>1000 but globally	Case fatality, recovery rate	Prevention, detection, response, risk environment indices and global health security (GHS)	Prevention (r=-0.988, -0.1.0 to -0.548), p= 0.012), detection (r=-0.979 (-1.000, -0.312), p=0.021), response (r=-0.965 (-0.999, -0.051), p=0.035), and GHS (r=-0.995 (-1.000, -0.786), P = 0.005) were negatively correlated with case fatality rate (CFR) in African countries.
Muneer *et al* (2020)[28]	Africa/Global	General	Web-based survey (worldometer)	-	Infection	Malaria prevalence	Ahigh rate of malaria was negatively correlated with Covid-19 (r= -0.15, p= 0.02).
Okpokoro et al (2020)[8]	Africa/Global	General	Ecologic study		Mortality	COPD, Tobacco use	Mortality was positively correlated with COPD (Rho=-0.28, p=0.09) and Tobacco use (Rho=-0.01, p=0.91) although not statistically significant. Similarly, the study showed that mortality was negatively correlated with life expectancy (Rho=-0.09, p=0.24),

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Author [30]	Setting	Population	Design	Sample (n)	COVID-19 Outcome	Exposure	Summary of study findings
							quality of air (Rho=-0.02, p=0.84), and life expectancy (Rho=-0.24, p=0.1).
Onovo et al (2020)[9]	Sub-Saharan Africa (SSA)	General	Survey	-	Infection	Adult HIV incidence rate, infant's pneumococcal conjugate-based vaccine, incidence of malaria, diarrhea treatment for under 5	HIV incidence (p= 0.0001), pneumococcal conjugate-based vaccine (p= 0.001), incidence of malaria (p= 0.001) and diarrhea treatment (p= 0.002) could significantly predict COVID-19 infection in sub-Saharan Africa (SSA). However, TB incidence (p= 0.86) and smoking prevalence (p= 0.53) were not statistically associated with coronavirus disease infection.
Ortiz-Fernández et al (2020)[29]	Africa/Global	General	Genetics	2,504 .	Susceptibility	Lower expression of angiotensin-converting enzyme 2 (ACE2) and type II transmembrane serine protease (TMPRSS2) genes	Africans were reported to have lower expression levels of ACE2 and TMPRSS2, suggesting a lower susceptibility for SARS-CoV-2 infection in the African populations.

CKD: chronic kidney disease; COPD: chronic obstructive pulmonary disease; CRD: chronic respiratory disease; CVD: cardiovascular diseases; DM= diabetes mellitus; GDP: gross domestic products; HTN= hypertension

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Factors affecting COVID-19 infection, severity, and related death

Demographic, institutional, economic, environmental, health system policy, lifestyle and political factors were reported to have an effect on prevention and spread of COVID-19 , so were the presence and level of chronic conditions [7-9 18-29]. For example, we found that COVID-19 infection and related deaths were affected by political democracy and economic openness as measured by volume of imports of goods and services and international tourism. Furthermore, high rate of comorbid conditions was positively correlated with COVID-19 infection and related death, with people having one or more chronic conditions experiencing severe form of the disease. A detailed summary of all the relevant factors affecting the infection, severity, and mortality related to COVID-19 is presented in Table 2.

Table 2. Factors associated with COVID-19 infection, severity and related death in Africa

Outcome	Theme	Potential specific factors
Infection	Demographic factors	High number of people aged ≥65 years[25], high population density[21 25]
	Health system level factors	Low number of physicians and hospitals[22], low BCG vaccination[24]
	Politico-economic factors	High economic openness, strong political democracy[23], more imports of goods and services, wide international tourism, high international travel[24], and city connectivity[20],
	Environmental factor	Low mean temperature[23 25]
	Comorbid conditions	Low malaria infection rate[9 25 28], high HIV infection rate and under-5 diarrheal disease[9]
Severity	Demographic factors	High percentage of older age population profile[7 19]
	Comorbid conditions	People with HIV, TB and anemia [19 21]
Death	Health system policy factors	Low prevention, detection, response and risk environment indices[27]
	Politico-economic factors	More imports of goods and services, wide international tourism[24], low global health security index score[27] and low health expenditure (percentage of GDP)[26]
	Chronic illness and lifestyle factors	High prevalence of chronic illness and Tobacco use[8]

BCG: Bacillus Calmette–Guérin; HIV: Human Immunodeficiency Virus; TB: Tuberculosis; GDP: Gross Domestic Product; COPD: Chronic Obstructive Pulmonary Disease

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Quality and risk of bias assessment

Detailed results of the methodological quality and risk of bias assessment of individual studies according to the JBI tool are reported in Appendix 4. In summary, 5(34%) studies scored 100%, 6 (40%) studies scored 80-89% and 3 (20%) studies scored 70-79%. Excluding studies with low scores did not materially change the results.

DISCUSSION

The systematic review presents a comprehensive summary of the current evidence on risk factors of COVID-19 infection, severity, and related deaths in Africa. Demographic, institutional, political, and ecological factors were linked with high COVID-19 infection rates in Africa. The study also revealed that severe forms of COVID-19 were associated with comorbidities and specific demographic characteristics. Health system organization and policy, politico-economic situation, prevalence of chronic conditions, and lifestyle factors increase the risk of deaths associated with COVID-19.

African countries which heavily rely on international trade and tourism were likely to have higher infection rates than other countries in the region [23 24]. As anticipated, the rate of COVID-19 infection was higher among densely populated countries given that the disease is mainly spread through close contact with infected persons [31 32]. A similar finding was observed elsewhere[33].

African countries enjoying a stable political democracy have an open economy and well connected air and road network were found to have a higher risk of COVID-19 infection [20 23]. This could be attributed to the fact that countries with strong democracy have a laissez-faire administrative system, where people are not obliged to obey and practice some public health measures that may infringe freedom of movement, and the pursuit of happiness. Democratic systems also go hand in hand with law and order and a higher than average disposable income, which create opportunities for leisure and work-related travels both internally and internationally, which increase the risk of COVID-19 transmission. In addition, countries with open economies and high transport connectivity are characterised by high mobility, facilitating spread of the disease.

The studies included in this review reported mixed findings with regards to the effect of temperature on COVID-19 infection. For example, the current review indicated that people living in countries with warmer temperature are less likely to acquire infection [23 34]. Contrary to this observation, some African countries such as Egypt, which enjoys warm weather

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through the year, are highly affected by the virus as compared to other African countries with relatively cooler temperature[35]. Yao and colleagues, on the other hand, [36] demonstrated that temperature had no significant association with COVID-19 infection. As highlighted in the study[33], further investigation is required to determine the role of weather on transmission and spread of COVID-19.

Age, particularly being an older person, was associated with more severe forms of COVID-19 at admission to hospital[37]. This might be explained by weaker immunity among older people. Patients with co-infections such as TB are more likely to experience severe forms of COVID-19 [21]. TB infection is common in low-resource settings and among older adults with pre-existing conditions resulting in high vulnerability to severe form of COVID-19 infection[19] [38]. Evidence on the effect of COVID-19 on people with HIV and malaria is limited [39]. A study by Karmen et al showed that being infected with HIV at the time of acquiring COVID-19 does not significantly increase the severity of illness or the risk of complications [40]. However, further research is warranted as the available evidence is inconclusive. Most importantly, given the high burden of HIV and malaria in the Africa region the molecular, genetic, clinical and environmental implications of COVID-19 on people living with HIV and malaria should be explored in greater detail.

Generally, patients with underlining chronic conditions are at a higher risk of having sever COVID-19 and related mortality. For instance, a study conducted by Cox et al[39] revealed greater risk and severe forms of COVID-19 in patients with COPD. Findings of a meta-analysis of seven studies conducted in China also reported significant association between COVID-19 and hypertension, chronic respiratory disease, and cardiovascular disease [41]. These findings imply that countries with a high burden of chronic conditions are more likely to have more severe cases of COVID-19. People who died of COVID-19 had chronic hypertension, COPD and cardiovascular comorbidities than recovered patients [42] [8]. History of tobacco use was found to be associated with increased risk of COVID-19 death [8] [43]. Additionally, Zhou and colleagues [44] found that 50% of patients who died of COVID-19 had a history of secondary bacterial infections. Chen and colleagues [45] also reported that bacterial and fungal co-infections increase the risk of COVID-19 related mortality. Thus, public health and medical services responding to the pandemic in Africa should be equipped with resources that enable the identification of cases with underlining chronic conditions and provide tailored interventions.

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The present review revealed that countries with effective health care system such as a strong disease prevention, case detection and response program and have a strong global health security system in pace had lower case fatality rate[27]. This was supported by a WHO report that the COVID-19 pandemic is straining health systems worldwide as health facilities have been overstretched and unable to operate effectively due to increased burden for health facilities and health care workers [46]. Importantly, higher per capita expenditure on health significantly reduces the risk of COVID-19 deaths [26]. This could be attributed to the fact that a strong health system with sufficient resources is both adaptive, resilient and often has the required manpower and facilities to respond to the higher demand for medical staff and supplies in treating COVID-19 cases.

The study has the following limitations. First, the results reported in the paper with respect to associated factors do not necessarily imply causality as most of the included studies were based on either cross-country or cross-sectional designs that are not suitable for studying causal relationships. Second, we have used hospital admission as one of the indicators of severity, but this may not be a sufficient criterion given that in some countries, especially in the early stage of the pandemic, all symptomatic patients were systematically hospitalized to avoid transmission. Third, included articles were limited to those written in English. Fourth, there is still limited evidence on COVID-19 in countries Africa (some of the included studies are even pre-print studies) and the review was unable to provide a stratified analysis by regions. Future research on COVID-19 in the region and beyond should focus on robust epidemiological study designs that are suitable to capture causal relationships and long-term impacts of the disease.

CONCLUSIONS

This systematic review demonstrated that several demographic, institutional, political, -economic, environmental, lifestyle and health system factors as well as comorbid conditions increased the risk infection, severe forms of COVId-19 and deaths related to the virus in the Africa region. The impacts of these factors should be factored in by African Governments and their development partners while designing tailored and targeted interventions to mitigate effects of the disease in the region. Implementing complete lock-down measures for an extended period in African countries is difficult or at least not as easy as in western countries, due to the region's unique social and economic settings. Alternative and effective measures

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such as mandatory face mask use and strong contact tracing system could be the most feasible interventions to contain the pandemic in the region [47]. However, it should be noted that even with a complete adherence to such public health measures, governments can only expect to mitigate the spread of the virus in the region. Eventually, safe and effective vaccines and drugs are required to end this pandemic.

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Figure Legends

Figure 1. PRISMA Flow Diagram— Figure 1 describes the flow diagram of the search strategy.

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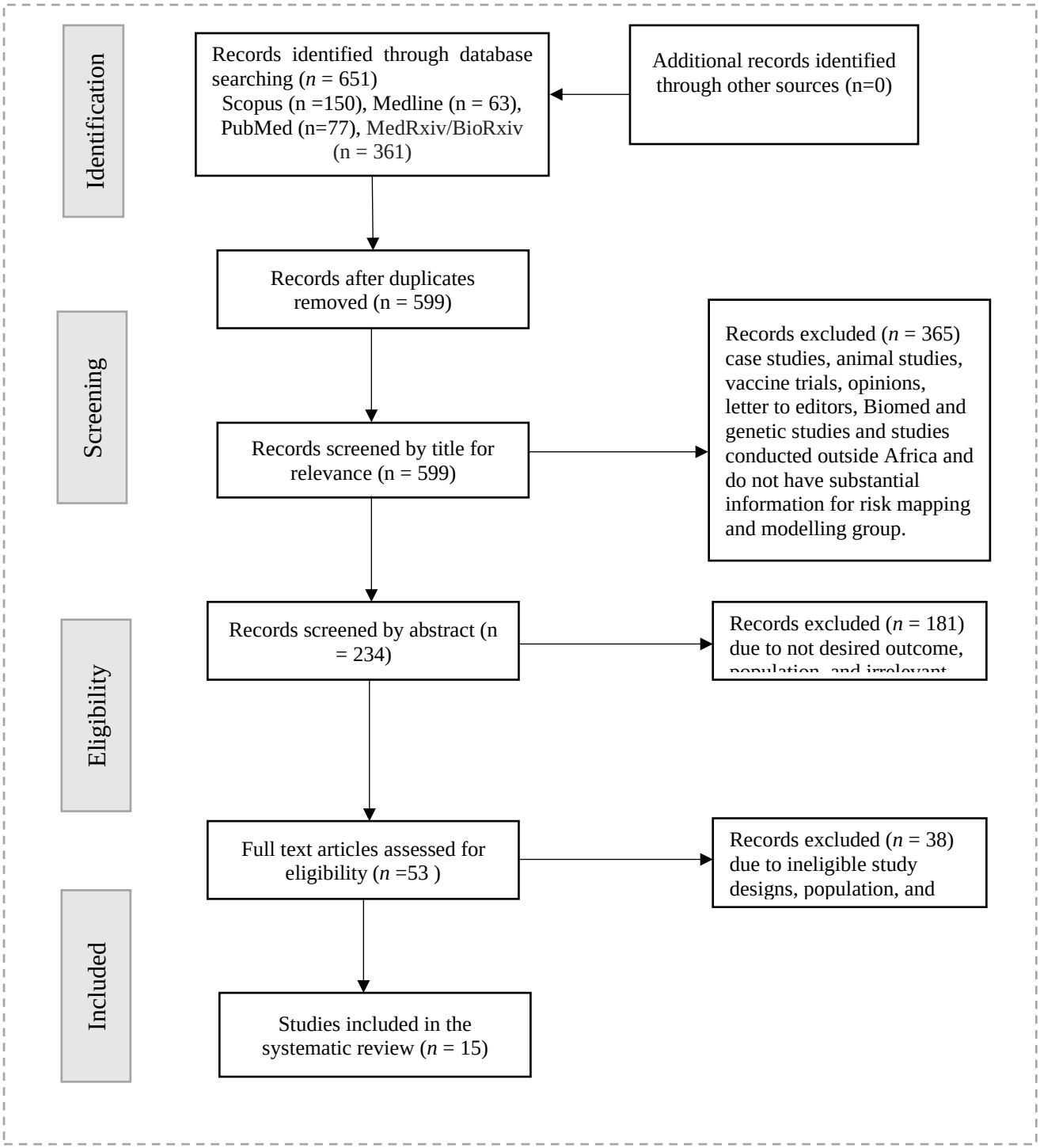


Figure 1. PRISMA Flow Diagram

Appendix 1: Search strategy

No	Terms/keywords
Medline	
1	Coronavirus/ or Coronavirus Infections/ or COVID-19.mp
2	coronavirus.mp
3	SARS-CoV-2.mp
4	n-CoV.mp
5	2019-nCoV.mp
6	1 or 2 or 3 or 4 or 5
7	Africa.mp. or exp Africa/
8	6 and 7
9	limit 15 to (English language and humans and year="2019 - 2020")
PubMed	
1	"COVID-19" OR coronavirus OR "SARS-CoV-2" OR "n-CoV" OR "2019-nCoV"
2	Africa*
3	(("COVID-19" OR coronavirus OR "SARS-CoV-2" OR "n-CoV" OR "2019-nCoV")) AND (Africa)
4	(("COVID-19" OR coronavirus OR "SARS-CoV-2" OR "n-CoV" OR "2019-nCoV")) AND (Africa) Sort by: Best Match Filters: Publication date from 2019/12/01; Humans; English
Scopus	(TITLE-ABS-KEY ("COVID-19" OR coronavirus OR "SARS-CoV-2" OR "n-CoV" OR "2019-nCoV")) AND (TITLE-ABS-KEY (africa) AND (LIMIT-TO (PUBYEAR , 2020) OR LIMIT-TO (PUBYEAR , 2019))

*list of countries was considered

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Appendix 2: JBI quality appraisal and selection tool

JBI Critical Appraisal Checklist for Descriptive / Case Series

Reviewer _____ Date _____

Author _____ Year _____ Record Number _____

	Yes	No	Unclear	Not Applicable
1. Was study based on a random or pseudo-random sample?	☐	☐	☐	☐
2. Were the criteria for inclusion in the sample clearly defined?	☐	☐	☐	☐
3. Were confounding factors identified and strategies to deal with them stated?	☐	☐	☐	☐
4. Were outcomes assessed using objective criteria?	☐	☐	☐	☐
5. If comparisons are being made, was there sufficient descriptions of the groups?	☐	☐	☐	☐
6. Was follow up carried out over a sufficient time period?	☐	☐	☐	☐
7. Were the outcomes of people who withdrew described and included in the analysis?	☐	☐	☐	☐
8. Were outcomes measured in a reliable way?	☐	☐	☐	☐
9. Was appropriate statistical analysis used?	☐	☐	☐	☐

Overall appraisal: Include ☐ Exclude ☐ Seek further info ☐

Comments (Including reason for exclusion)

Appendix 3: JBI data extraction tool

JBI Data Extraction Form for Experimental / Observational Studies

Reviewer _____ Date _____

Author _____ Year _____

Journal _____ Record Number _____

Study Method

RCT ☐ Quasi-RCT ☐ Longitudinal ☐

Retrospective ☐ Observational ☐ Other ☐

Participants

Setting _____

Population _____

Sample size

Group A _____ Group B _____

Interventions

Intervention A _____

_____

Intervention B _____

_____

Authors Conclusions:

_____

_____

_____

Reviewers Conclusions:

_____

_____

_____

Appendix 4: Assessment of methodological quality (n=15)

Authors, year	*Cross-sectional studies*									
	Was study based on a random or pseudo-random sample?	Were the criteria for inclusion in the sample clearly defined?	Were confounding factors identified and strategies to deal with them stated?	Were outcomes assessed using objective criteria?	If comparisons are being made, was there sufficient comparison of groups?	Was the follow-up carried out over a sufficient period?	Were the outcomes of people who withdrew described and included in the analysis?	Were outcomes measured in reliable way?	Was appropriate statistical analysis used?	%
Bergman et al (2020)[18]	N	NA	Y	Y	NA	Y	NA	U	Y	80
Clark et al (2020)[19]	NA	Y	Y	Y	NA	NA	NA	Y	Y	100
Daon et al(2020)[20]	NA	Y	Y	Y	NA	NA	NA	Y	Y	100
Davies et al (2020)[7]	NA	Y	Y	Y	NA	NA	NA	Y	Y	100
Diop et al (2020)[21]	NA	Y	Y	Y	NA	NA	NA	Y	Y	100
Hossain et al (2020)[23]	N	Y	U	Y	Y	N	NA	Y	Y	72
Jaffé et al (2020)[24]	N	Y	U	Y	Y	N	NA	Y	Y	72
Kubota *et al* (2020)[25]	N	Y	Y	Y	Y	N	NA	Y	Y	86
Maitra et al (2020)[26]	N	Y	Y	Y	NA	NA	NA	Y	Y	83
Gayawan et al (2020)[22]	N	Y	N	Y	Y	N/A	N/A	Y	Y	80%
Maraghi et al. (2020)[27]	Yes	Yes	Yes	No	Yes	NA	NA	No	Yes	85.7
Muneer *et al* (2020)[28]	Y	Y	N	Y	Y	NA	NA	Y	Y	86
Okpokoro et al (2020)[8]	Y	Y	N	Y	Y	NA	NA	Y	Y	86
Onovo et al (2020)[9]	Y	Y	Y	Y	Y	NA	NA	Y	Y	100
Ortiz-Fernández et al (2020)[29]	Y	Y	N	Y	N	NA	NA	Y	Y	72

Y=Yes; N= No; U= Unclear; NA= Not applicable; %= Percentage of score

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Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	3
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	5-6
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	6
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	No protocol
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	7
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	7
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	7
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	7
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	8
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	7-8
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	8
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	8
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	8

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Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	8
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	Not Applicable
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	9
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	9
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	9, and Appendix 4
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	10-13
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	Not Applicable
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	Not Applicable
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	15-16
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	17
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	17
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	17

From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(6): e1000097. doi:10.1371/journal.pmed1000097

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Risk factors for COVID-19 infection, disease severity and related deaths in Africa: a systematic review

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Primary Subject Heading:	Epidemiology
Secondary Subject Heading:	Infectious diseases
Keywords:	Public health < INFECTIOUS DISEASES, EPIDEMIOLOGY, RESPIRATORY MEDICINE (see Thoracic Medicine)

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Risk factors for COVID-19 infection, disease severity and related deaths in Africa: a systematic review

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ABSTRACT

Objective: The aim of this study was to provide a comprehensive evidence on risk factors for transmission, disease severity, and COVID-19 related deaths in Africa.

Design: A systematic review has been conducted to synthesise existing evidence on risk factors affecting COVID-19 outcomes across Africa.

Data sources: Data were systematically searched from MEDLINE, Scopus, MedRxiv and BioRxiv.

Eligibility criteria: Studies for review were included if they were published in English and reported at least one risk factor and/ or one health outcome. We included all relevant literature published up until 11 August 2020.

Data extraction and synthesis: We performed a systematic narrative synthesis to describe the available studies for each outcome. Data were extracted using a standardised JBI data extraction form.

Results: Fifteen articles met the inclusion criteria of which four were exclusively on Africa and the remaining 11 papers had a global focus with some data from Africa. Higher rates of infection in Africa are associated with high population density, urbanization, transport connectivity, high volume of tourism and international trade, and high level of economic and political openness. Limited or poor access to health care are also associated with higher COVID-19 infection rates. Older people and individuals with chronic conditions such as human immunodeficiency virus (HIV), tuberculosis and anaemia experience severe forms COVID-19 leading to hospitalisation and death. Similarly, high burden of chronic obstructive pulmonary disease (COPD), high prevalence of tobacco consumption, and low levels of expenditure on health and low levels of global health security score contribute to COVID-19 related deaths.

Conclusions: Demographic, institutional, ecologic, health system and politico-economic factors influenced the spectrum of COVID-19 infection, severity and death. We recommend multidisciplinary and integrated approaches to mitigate the identified factors and strengthen effective prevention strategies.

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Key words: ABReN, Africa, COVID-19, death, infection, risk factors, severity, systematic review

Strengths and limitations

- Effective public health interventions against the COVID-19 pandemic require strong evidence on the risk factors associated with the disease.
- The present study is the first African-focused comprehensive systematic review of its kind on a wide range of risk factors associated with the entire spectrum of COVID-19 in the Africa region.
- Titles and abstracts of all publications identified by the search criteria were screened by two independent reviewers of the team.
- Full text articles were appraised using the Joanna Briggs Institute critical appraisal tools.
- The available data did not allow us to carry out meta-analysis nor firmly establish causal relationships in the present study.

INTRODUCTION

The coronavirus disease (COVID-19) is a recently emerged viral disease caused by a novel single-stranded enveloped RNA (ribonucleic acid) virus, known as Severe Acute Respiratory Syndrome Coronavirus-2 (SARS-CoV-2)[1]. Since the first COVID-19 case was identified in Wuhan province in China, in late December 2019, the virus has rapidly spread across the world and declared a global pandemic on 11 March 2020[2]. As of 10 January 2021, the number of confirmed COVID-19 cases in the world has reached over 89.44 million with nearly 1.92 million deaths[3]. Despite the late start of the pandemic, infection rate in Africa has been increasing exponentially with a total of 2.95 million confirmed cases and more than 70 thousand deaths as of 10 January 2021[4]. The majority of reported cases in Africa were from South Africa and Egypt, while countries in Central Africa had recorded the least.

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Given that the pandemic is caused by a novel strain of coronavirus with unknown original host[5], in the early stage of the outbreak the risk factors associated with its transmission routes, severity, and fatality risks were unclear. Several studies have been conducted to better understand the prognosis of the disease, though the level of uncertainty on viral shedding, transmissibility and disease severity remains high[6]. Evidence indicates that the susceptibility to infection, being seriously ill and the risk of death are influenced by individual-level characteristics such as socio-demographic factors[7], behavioural traits[8], and pre-existing medical conditions[9].

Whilst the number of people affected by the virus in Africa remains to be the lowest by global standards, which in part may be linked to the youthful age structure of the population and partly to the limited capacity for large scale testing[10], it is anticipated that the pandemic may have a profound impact in the region. First and foremost, Africa's health system is fragile. In addition, preventive strategies such as social distancing and hand washing, which have proved effective in reducing transmission in the rest of the world, are less likely to be practical in the region due to prevailing socio-cultural and economic circumstances[11]. Compared to other regions, Africa is home to a large number of people living in sub-standard housing, which are mostly dense, inadequately ventilated and have limited access to direct water supply. The way of life in the region is largely communal. People tend to attend weddings, religious and funeral services in large numbers. Such societal and environmental factors along with the lack of awareness and access to preventive measures, and the potential for low level of adherence (even when the resources are available) mean that the region will have the highest number of vulnerable populations for COVID-19 infection [12].

Many countries have undertaken strict measures such as banning of public gatherings, complete lock-down of social and economic activities, and closure of borders to prevent importation of cases[13]. The number of new cases, however, continues to rise[11] creating considerable pressure on the health-care system[14]. In the absence of effective vaccine or therapy, the ability to effectively control the spread of the pandemic, hence depends on effective monitoring of new cases, and on better understanding of the factors

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related to modes of transmission and severity of the disease. A comprehensive systematic review of African-focused COVID-19 related studies is required to strengthen public health measures and response plan against the pandemic in the continent. This review aims to provide a comprehensive evidence on risk factors for transmission, disease severity, and COVID-19 related deaths in Africa.

METHODS

Australia Based Ethiopian Researchers Network (ABReN)

At the beginning of April, 2020, a time when the COVID-19 pandemic was spreading across the globe, a group of more than 50 Australian based Ethiopian researchers, who are active academic faculties in Australian institutions along with Ethiopian PHD students, established a working group on "Ethiopia and the COVID-19 pandemic" to examine various dimensions and consequences of COVID-19 in Ethiopia. Specifically, the working group focussed on four distinct thematic areas: (i) on understanding the epidemiology of COVID-19 in Ethiopia and around the globe and synthesising the lessons for the country, (ii) assessing opportunities for re-imagining the health system for COVID-19 pandemic response, (iii) identifying potential pharmaceutical and non-pharmaceutical interventions, and (iv) measuring socio-economic impact of COVID-19 in Ethiopia to support the Ethiopian government's effort to mitigate the deadly impact of the pandemic on the population. As part of the project under the first thematic area, we conducted a systematic review to synthesise the available evidence on COVID-19 in Africa on factors affecting infection rates, severity, and related deaths.

Search strategy

This systematic review is reported according to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses guidelines [15]. Potentially eligible articles were systematically searched and retrieved from Medline, PubMed and Scopus up to 11 August 2020. The following search terms, tailored to each database were used: "COVID-19" OR "coronavirus disease" OR "SARS-COV-2" OR "2019 novel coronavirus" OR "2019-nCoV" AND "Africa". The full search strategy is summarised in Appendix 1. We used medical

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subject headings (MeSH) terms to identify synonyms and text words in the appropriate syntax of each database. The search was not limited by date of publication or study design. Reference lists of included studies were manually searched for additional relevant articles. Moreover, preprint articles from the MedRxiv and BioRxiv databases were also accessed to ensure wider coverage. However, search was restricted to studies conducted in Africa and the English language only.

Study selection criteria

The titles and abstracts of all publications identified by the search criteria were screened by two independent reviewers of the team (AAA and BAD) to retrieve full texts for all relevant studies. Full text articles were appraised using the Joanna Briggs Institute [16] critical appraisal tools[17] (Appendix 2). All studies published in English and reported at least one risk factor and/ or one health outcome were included in the review. We excluded letters to editors, correspondence, editorials, and case studies. However, we included global studies which reported findings from any country in Africa.

Socio-demographic, lifestyle and behavioural, climate variables and chronic comorbid conditions, which can be at a community or national level, were considered as risk factors. Whereas COVID-19 infection rates, level of severity, and COVID-19 related deaths were considered as outcomes. We adopted the definition of terms as reported in each of the included studies. Severity was referred for cases which required hospitalisation and are at high risk of death.

Data extraction

Seven members of the research team (HAG, DNK, DMF, MW, ATG, BAD and AAA) extracted needed data using a standardised JBI data extraction form[17] (Appendix 3). The extracted data included study authors, setting, population characteristics, study design, outcome, and main findings on one or more risk factors and reported measure of association between risk factors and outcomes.

Quality and risk of bias assessment in individual studies

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The JBI tools[17] were used to assess the methodological quality and risk of bias (in design, conduct and analysis) of included studies (Appendix 2). The tool is used to assess the inclusion criteria, measurement of exposure and outcome variables, confounding adjustment, and appropriateness of statistical analysis. All authors who participated in the data extraction also assessed the quality of studies that they extracted.

Data synthesis

A systematic narrative synthesis was conducted to describe the available studies separately for each outcome. Five steps were applied in synthesizing the data: (i) reading findings and results of each study systematically and comprehensively and assessing for relevance; (ii) developing a preliminary synthesis through describing each included study, grouping findings by infection, severity and death and tabulate results to identify patterns; (iii) organize factors linked to the outcomes described in phase (ii) and thematize items around themes; and (iv) explore the relationships of each factor within and among themes. Meta-analysis was not conducted because of the small number of studies identified for selected risk factors.

Patient and public involvement

Patients and/or the public were not involved in the design, or conduct, or reporting, or dissemination plans of this research.

RESULTS

Description of studies

The combined search terms identified 651 records (Figure 1). After removing duplicates (n=52) and unrelated studies (n=584), 53 records were eligible for full text screening. Finally, 15 studies [7-9 18-29] met the inclusion criteria and were included in this systematic review. Table 1 presents the main characteristics and outcomes of these studies. Four studies were exclusively on Africa[9 18 21 22]. Two studies[7 29] assessed susceptibility to COVID-19 infection, nine studies[7 9 18 20 22-25 28] evaluated COVID-19 infection, three studies[7 19 21] examined severity of COVID-19 infection, and four studies[8 24 26 27] assessed COVID-19 related death. Four studies assessed more than one outcome[7 21 24 27]. As described in Table 1, most studies (13) were conducted among the general population. Study designs

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such as modelling (n=7), cross-sectional (n=2), genetics (n=1), survey (n=4) and Spatio-temporal analysis (n=1) were used in the included studies. The number of participants of those included studies for review ranges from 446 to 1.7 billion participants—the studies with the largest sample size were modelling related studies. We have annexed the quality score of all included studies in Appendix 4.

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Table 1. Characteristics of studies (n=15 studies) included in the systematic review

Author	Setting	Population	Design	Sample (n)	COVID-19 Outcome	Exposure	Summary of study findings
Bergman et al (2020)[18]	Africa	Adults	Modelling	446	Infection	Mobility	There was no statistically significant association between transmission rates and lagged mobility. But there is limited information about Google workplace mobility data and COVID-19 infection as Apple mobility data is limited in Africa unlike other regions.
Clark et al (2020)[19]	Africa/Global	General	Modelling	1.7 Billion	Severity	Age, sex, comorbidity and continent	Individuals who are diagnosed with at least one underlying condition are at risk of severe COVID-19 disease. Although not different by sex, a total of 1.17 billion (22%) of the global population were estimated to be at risk of sever COVID-19 disease, and the prevalence of one or more co-moribid condition was approximately 10% by age 25 years, 33% by 50 years, and 66% by 70 years. A total of 400 million (6%) of the global population were estimated to be at risk of sever COVID-19 disease, and is higher in older people. The most prevalent conditions in those aged 50+ years were CKD, CVD, CRD and diabetes— the prevalence of multimorbidity (2+ underlying chronic illness) was three times higher in Europe than in Africa (10% vs 3%).
Daon et al(2020)[20]	Africa/Global	General	Modelling	1364 major airports (connections)	Infection	City connectivity	An outbreak of COVID-19 in Africa is most likely to originate from a passenger who travelled from Europe, and the average risk of COVID-19 infection originated from airport in Africa is 0.06, lower result compared to Airports in Asia with an estimated risk of 0.51). The airport with the highest risk to initiate an outbreak in Africa is in Johannesburg with a risk estimate of 0.1.
Davies et al (2020)[7]	Africa/Global	General	Modelling	-	Infection, Severity	Age	People younger than 20 years old are roughly half susceptible as those who are older than 20 years old. In addition, 75% of infections are subclinical in

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Author	Setting	Population	Design	Sample (n)	COVID-19 Outcome	Exposure	Summary of study findings
							children ageing between 10 and19 years as compared to 24% in people aged 70 years and above.
Diop et al (2020)[21]	Africa	General	Modelling	-	Infection Severity	Residence, population density, comorbidity (HIV, Tb and Anemia)	The rate of COVID-19 infection increases as the population density increases. Linked to this, the rate of infection is low in rural settings than in urban areas. The study reported that comorbidity would increase severity of the disease. The rate of anemia, TB & HIV compared to the rate of DM, HTN and obesity is higher in Africa than other continents, and these (anemia, TB and HIV) are the underlying conditions which lead to COVID-19 related severity in Africa.
Gayawan et al (2020)[22]	Africa	General	Spatio-temporal analysis	Not specified	Infection	number of physicians number of hospital beds	Positive correlation was revealed between COVID-19 and number of physicians (r=0.49, p-value=<0.001) and hospital beds (r=0.14, p-value=0.34) though only the estimate for physicians is significant. The Spatio-temporal analysis reveals that the occurrence and burden of COVID-19 in Africa varied geographically with neighboring countries particularly in the western part of the continent which could imply that neighboring countries pose significant importation risk to each other
Hossain et al (2020)[23]	Africa/Global	General	Modelling	1,174,652	Infection	Weather, economic openness, and political democracy	Countries with warmer temperatures showed slower spread than their comparator. Countries with higher economic openness had higher infection rates than with lower economic openness. The countries with stronger political democracy had higher infection rates than with weaker political democracy.
Jaffé et al (2020)[24]	Africa/Global	General	Cross-sectional	Dataset of 36 countries	Infection Mortality	imports of goods and services, international tourism, electric power consumption,	Countries that were connected to the global economy tended to have a higher risk of infection, while the risk of mortality from the disease are higher in less globalized low-income countries. Countries with higher imports of goods and services, and

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Author	Setting	Population	Design	Sample (n)	COVID-19 Outcome	Exposure	Summary of study findings
						and Population over 65 years old	international tourism had higher infection rates than their comparator.
Kubota *et al* (2020)[25]	Africa/Global	General	Modelling study	1,055 countries	Infection	Age, precipitation, temperature, international travel, BCG vaccination, malaria infection	A negative correlation between the accumulated numbers of the COVID-19 cases and the following variables: mean temperature, BCG vaccination effect and malaria infection. The study found a positive correlation between the accumulated numbers of the COVID-19 cases and the following variables: mean precipitation, the relative frequency of foreign visitors per population, GDP per person, population density, and relative proportion of people ≥ 65 years old.
Maitra et al (2020)[26]	Africa/Global	General	Cross sectional	422,582	Mortality	Health expenditure	Countries with low health expenditure (percentage of GDP) were significantly associated with higher case fatality rate (p=0.0017).
Maraghi et al. (2020)[27]	Africa/Global	Not specified	Cross-sectional	>1000 but globally	Case fatality, recovery rate	Prevention, detection, response, risk environment indices and global health security (GHS)	Prevention (r=-0.988, -0.1.0 to -0.548), p= 0.012), detection (r=-0.979 (-1.000, -0.312), p=0.021), response (r=-0.965 (-0.999, -0.051), p=0.035), and GHS (r=-0.995 (-1.000, -0.786), P = 0.005) were negatively correlated with case fatality rate (CFR) in African countries.
Muneer *et al* (2020)[28]	Africa/Global	General	Web-based survey (worldometer)	-	Infection	Malaria prevalence	Ahigh rate of malaria was negatively correlated with Covid-19 (r= -0.15, p= 0.02).
Okpokoro et al (2020)[8]	Africa/Global	General	Ecologic study		Mortality	COPD, Tobacco use	Mortality was positively correlated with COPD (Rho=-0.28, p=0.09) and Tobacco use (Rho=-0.01, p=0.91) although not statistically significant. Similarly, the study showed that mortality was negatively correlated with life expectancy (Rho=-0.09, p=0.24),

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Author	Setting	Population	Design	Sample (n)	COVID-19 Outcome	Exposure	Summary of study findings
							quality of air (Rho=-0.02, p=0.84), and life expectancy (Rho=-0.24, p=0.1).
Onovo et al (2020)[9]	Sub-Saharan Africa (SSA)	General	Survey	-	Infection	Adult HIV incidence rate, infant's pneumococcal conjugate-based vaccine, incidence of malaria, diarrhea treatment for under 5	HIV incidence (p= 0.0001), pneumococcal conjugate-based vaccine (p= 0.001), incidence of malaria (p= 0.001) and diarrhea treatment (p= 0.002) could significantly predict COVID-19 infection in sub-Saharan Africa (SSA). However, TB incidence (p= 0.86) and smoking prevalence (p= 0.53) were not statistically associated with coronavirus disease infection.
Ortiz-Fernández et al (2020)[29]	Africa/Global	General	Genetics	2,504 .	Susceptibility	Lower expression of angiotensin-converting enzyme 2 (ACE2) and type II transmembrane serine protease (TMPRSS2) genes	Africans were reported to have lower expression levels of ACE2 and TMPRSS2, suggesting a lower susceptibility for SARS-CoV-2 infection in the African populations.

CKD: chronic kidney disease; COPD: chronic obstructive pulmonary disease; CRD: chronic respiratory disease; CVD: cardiovascular diseases; DM= diabetes mellitus; GDP: gross domestic products; HTN= hypertension

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Factors affecting COVID-19 infection, severity, and related death

Demographic, institutional, economic, environmental, health system policy, lifestyle and political factors were reported to have an effect on prevention and spread of COVID-19 , so were the presence and level of chronic conditions [7-9 18-29]. For example, we found that COVID-19 infection and related deaths were affected by political democracy and economic openness as measured by volume of imports of goods and services and international tourism. Furthermore, high rate of comorbid conditions was positively correlated with COVID-19 infection and related death, with people having one or more chronic conditions experiencing severe form of the disease. A detailed summary of all the relevant factors affecting the infection, severity, and mortality related to COVID-19 is presented in Table 2.

Table 2. Factors associated with COVID-19 infection, severity and related death in Africa

Outcome	Theme	Potential specific factors
Infection	Demographic factors	High number of people aged ≥65 years[25], high population density[21 25]
	Health system level factors	Low number of physicians and hospitals[22], low BCG vaccination[24]
	Politico-economic factors	High economic openness, strong political democracy[23], more imports of goods and services, wide international tourism, high international travel[24], and city connectivity[20],
	Environmental factor	Low mean temperature[23 25]
	Comorbid conditions	Low malaria infection rate[9 25 28], high HIV infection rate and under-5 diarrheal disease[9]
Severity	Demographic factors	High percentage of older age population profile[7 19]
	Comorbid conditions	People with HIV, TB and anemia [19 21]
Death	Health system policy factors	Low prevention, detection, response and risk environment indices[27]
	Politico-economic factors	More imports of goods and services, wide international tourism[24], low global health security index score[27] and low health expenditure (percentage of GDP)[26]
	Chronic illness and lifestyle factors	High prevalence of chronic illness and Tobacco use[8]

BCG: Bacillus Calmette–Guérin; HIV: Human Immunodeficiency Virus; TB: Tuberculosis; GDP: Gross Domestic Product; COPD: Chronic Obstructive Pulmonary Disease

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Quality and risk of bias assessment

Detailed results of the methodological quality and risk of bias assessment of individual studies according to the JBI tool are reported in Appendix 4. In summary, 5(34%) studies scored 100%, 6 (40%) studies scored 80-89% and 3 (20%) studies scored 70-79%. Excluding studies with low scores did not materially change the results.

DISCUSSION

The systematic review presents a comprehensive summary of the current evidence on risk factors of COVID-19 infection, severity, and related deaths in Africa. Demographic, institutional, political, and ecological factors were linked with high COVID-19 infection rates in Africa. The study also revealed that severe forms of COVID-19 were associated with comorbidities and specific demographic characteristics. Health system organization and policy, politico-economic situation, prevalence of chronic conditions, and lifestyle factors increase the risk of deaths associated with COVID-19.

African countries which heavily rely on international trade and tourism were likely to have higher infection rates than other countries in the region [23 24]. As anticipated, the rate of COVID-19 infection was higher among densely populated countries given that the disease is mainly spread through close contact with infected persons [30 31]. A similar finding was observed elsewhere[32].

African countries enjoying a stable political democracy have an open economy and well connected air and road network were found to have a higher risk of COVID-19 infection [20 23]. This could be attributed to the fact that countries with strong democracy have a laissez-faire administrative system, where people are not obliged to obey and practice some public health measures that may infringe freedom of movement, and the pursuit of happiness. Democratic systems also go hand in hand with law and order and a higher than average disposable income, which create opportunities for leisure and work-related travels both internally and internationally, which increase the risk of COVID-19 transmission. In addition, countries with open economies and high transport connectivity are characterised by high mobility, facilitating spread of the disease.

The studies included in this review reported mixed findings with regards to the effect of temperature on COVID-19 infection. For example, the current review indicated that people living in countries with warmer temperature are less likely to acquire infection [23 33]. Contrary to this observation, some African countries such as Egypt, which enjoys warm weather

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through the year, are highly affected by the virus as compared to other African countries with relatively cooler temperature[34]. Yao and colleagues, on the other hand, [35] demonstrated that temperature had no significant association with COVID-19 infection. As highlighted in the study[32], further investigation is required to determine the role of weather on transmission and spread of COVID-19.

Age, particularly being an older person, was associated with more severe forms of COVID-19 at admission to hospital[36]. This might be explained by weaker immunity among older people. Patients with co-infections such as TB are more likely to experience severe forms of COVID-19 [21]. TB infection is common in low-resource settings and among older adults with pre-existing conditions resulting in high vulnerability to severe form of COVID-19 infection[19] [37]. Evidence on the effect of COVID-19 on people with HIV and malaria is limited [38]. A study by Karmen et al showed that being infected with HIV at the time of acquiring COVID-19 does not significantly increase the severity of illness or the risk of complications [39]. However, further research is warranted as the available evidence is inconclusive. Most importantly, given the high burden of HIV and malaria in the Africa region the molecular, genetic, clinical and environmental implications of COVID-19 on people living with HIV and malaria should be explored in greater detail.

Generally, patients with underlining chronic conditions are at a higher risk of having sever COVID-19 and related mortality. For instance, a study conducted by Cox et al[38] revealed greater risk and severe forms of COVID-19 in patients with COPD. Findings of a meta-analysis of seven studies conducted in China also reported significant association between COVID-19 and hypertension, chronic respiratory disease, and cardiovascular disease [40]. These findings imply that countries with a high burden of chronic conditions are more likely to have more severe cases of COVID-19. People who died of COVID-19 had chronic hypertension, COPD and cardiovascular comorbidities than recovered patients [41] [8]. History of tobacco use was found to be associated with increased risk of COVID-19 death [8] [42]. Additionally, Zhou and colleagues [43] found that 50% of patients who died of COVID-19 had a history of secondary bacterial infections. Chen and colleagues [44] also reported that bacterial and fungal co-infections increase the risk of COVID-19 related mortality. Thus, public health and medical services responding to the pandemic in Africa should be equipped with resources that enable the identification of cases with underlining chronic conditions and provide tailored interventions.

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The present review revealed that countries with effective health care system such as a strong disease prevention, case detection and response program and have a strong global health security system in pace had lower case fatality rate[27]. This was supported by a WHO report that the COVID-19 pandemic is straining health systems worldwide as health facilities have been overstretched and unable to operate effectively due to increased burden for health facilities and health care workers [45]. Importantly, higher per capita expenditure on health significantly reduces the risk of COVID-19 deaths [26]. This could be attributed to the fact that a strong health system with sufficient resources is both adaptive, resilient and often has the required manpower and facilities to respond to the higher demand for medical staff and supplies in treating COVID-19 cases.

The study has the following limitations. First, the results reported in the paper with respect to associated factors do not necessarily imply causality as most of the included studies were based on either cross-country or cross-sectional designs that are not suitable for studying causal relationships. Second, we have used hospital admission as one of the indicators of severity, but this may not be a sufficient criterion given that in some countries, especially in the early stage of the pandemic, all symptomatic patients were systematically hospitalized to avoid transmission. Third, included articles were limited to those written in English. Fourth, there is still limited evidence on COVID-19 in countries Africa (some of the included studies are even pre-print and others are modelling based studies) and the review was unable to provide a stratified analysis by regions—authors of the modelling studies have already noted some limitations of the modelling studies. Future research on COVID-19 in the region and beyond should focus on robust epidemiological study designs that are suitable to capture causal relationships and long-term impacts of the disease.

CONCLUSIONS

This systematic review demonstrated that several demographic, institutional, political, -economic, environmental, lifestyle and health system factors as well as comorbid conditions increased the risk infection, severe forms of COVId-19 and deaths related to the virus in the Africa region. The impacts of these factors should be factored in by African Governments and their development partners while designing tailored and targeted interventions to mitigate effects of the disease in the region. Implementing complete lock-down measures for an extended period in African countries is difficult or at least not as easy as in western countries,

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due to the region's unique social and economic settings. Alternative and effective measures such as mandatory face mask use and strong contact tracing system could be the most feasible interventions to contain the pandemic in the region [46]. However, it should be noted that even with a complete adherence to such public health measures, governments can only expect to mitigate the spread of the virus in the region. Eventually, safe and effective vaccines and drugs are required to end this pandemic.

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Figure Legends

Figure 1. PRISMA Flow Diagram— Figure 1 describes the flow diagram of the search strategy.

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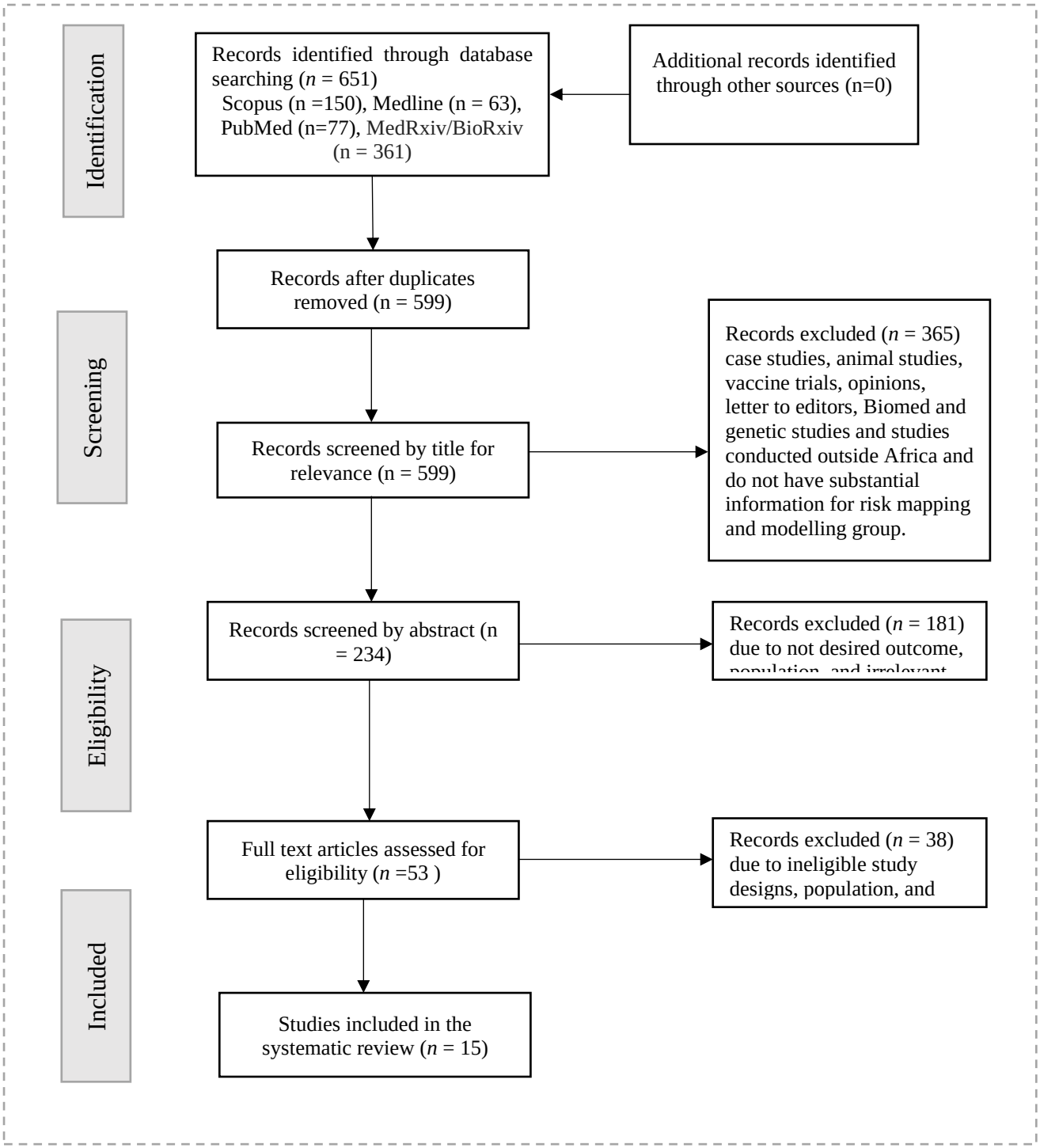


Figure 1. PRISMA Flow Diagram

Appendix 1: Search strategy

No	Terms/keywords
Medline	
1	Coronavirus/ or Coronavirus Infections/ or COVID-19.mp
2	coronavirus.mp
3	SARS-CoV-2.mp
4	n-CoV.mp
5	2019-nCoV.mp
6	1 or 2 or 3 or 4 or 5
7	Africa.mp. or exp Africa/
8	6 and 7
9	limit 15 to (English language and humans and year="2019 - 2020")
PubMed	
1	"COVID-19" OR coronavirus OR "SARS-CoV-2" OR "n-CoV" OR "2019-nCoV"
2	Africa*
3	(("COVID-19" OR coronavirus OR "SARS-CoV-2" OR "n-CoV" OR "2019-nCoV")) AND (Africa)
4	(("COVID-19" OR coronavirus OR "SARS-CoV-2" OR "n-CoV" OR "2019-nCoV")) AND (Africa) Sort by: Best Match Filters: Publication date from 2019/12/01; Humans; English
Scopus	(TITLE-ABS-KEY ("COVID-19" OR coronavirus OR "SARS-CoV-2" OR "n-CoV" OR "2019-nCoV")) AND (TITLE-ABS-KEY (africa) AND (LIMIT-TO (PUBYEAR , 2020) OR LIMIT-TO (PUBYEAR , 2019))

*list of countries was considered

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Appendix 2: JBI quality appraisal and selection tool

JBI Critical Appraisal Checklist for Descriptive / Case Series

Reviewer _____ Date _____

Author _____ Year _____ Record Number _____

	Yes	No	Unclear	Not Applicable
1. Was study based on a random or pseudo-random sample?	☐	☐	☐	☐
2. Were the criteria for inclusion in the sample clearly defined?	☐	☐	☐	☐
3. Were confounding factors identified and strategies to deal with them stated?	☐	☐	☐	☐
4. Were outcomes assessed using objective criteria?	☐	☐	☐	☐
5. If comparisons are being made, was there sufficient descriptions of the groups?	☐	☐	☐	☐
6. Was follow up carried out over a sufficient time period?	☐	☐	☐	☐
7. Were the outcomes of people who withdrew described and included in the analysis?	☐	☐	☐	☐
8. Were outcomes measured in a reliable way?	☐	☐	☐	☐
9. Was appropriate statistical analysis used?	☐	☐	☐	☐

Overall appraisal: Include ☐ Exclude ☐ Seek further info ☐

Comments (Including reason for exclusion)

Appendix 3: JBI data extraction tool

JBI Data Extraction Form for Experimental / Observational Studies

Reviewer _____ Date _____

Author _____ Year _____

Journal _____ Record Number _____

Study Method

RCT ☐ Quasi-RCT ☐ Longitudinal ☐

Retrospective ☐ Observational ☐ Other ☐

Participants

Setting _____

Population _____

Sample size

Group A _____ Group B _____

Interventions

Intervention A _____

_____

Intervention B _____

_____

Authors Conclusions:

_____

_____

_____

Reviewers Conclusions:

_____

_____

_____

Appendix 4: Assessment of methodological quality (n=15)

Authors, year	*Cross-sectional studies*									
	Was study based on a random or pseudo-random sample?	Were the criteria for inclusion in the sample clearly defined?	Were confounding factors identified and strategies to deal with them stated?	Were outcomes assessed using objective criteria?	If comparisons are being made, was there sufficient comparison of groups?	Was the follow-up carried out over a sufficient period?	Were the outcomes of people who withdrew described and included in the analysis?	Were outcomes measured in reliable way?	Was appropriate statistical analysis used?	%
Bergman et al (2020)[18]	N	NA	Y	Y	NA	Y	NA	U	Y	80
Clark et al (2020)[19]	NA	Y	Y	Y	NA	NA	NA	Y	Y	100
Daon et al(2020)[20]	NA	Y	Y	Y	NA	NA	NA	Y	Y	100
Davies et al (2020)[7]	NA	Y	Y	Y	NA	NA	NA	Y	Y	100
Diop et al (2020)[21]	NA	Y	Y	Y	NA	NA	NA	Y	Y	100
Hossain et al (2020)[23]	N	Y	U	Y	Y	N	NA	Y	Y	72
Jaffé et al (2020)[24]	N	Y	U	Y	Y	N	NA	Y	Y	72
Kubota *et al* (2020)[25]	N	Y	Y	Y	Y	N	NA	Y	Y	86
Maitra et al (2020)[26]	N	Y	Y	Y	NA	NA	NA	Y	Y	83
Gayawan et al (2020)[22]	N	Y	N	Y	Y	N/A	N/A	Y	Y	80%
Maraghi et al. (2020)[27]	Yes	Yes	Yes	No	Yes	NA	NA	No	Yes	85.7
Muneer *et al* (2020)[28]	Y	Y	N	Y	Y	NA	NA	Y	Y	86
Okpokoro et al (2020)[8]	Y	Y	N	Y	Y	NA	NA	Y	Y	86
Onovo et al (2020)[9]	Y	Y	Y	Y	Y	NA	NA	Y	Y	100
Ortiz-Fernández et al (2020)[29]	Y	Y	N	Y	N	NA	NA	Y	Y	72

Y=Yes; N= No; U= Unclear; NA= Not applicable; %= Percentage of score

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Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	3
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	5-6
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	6
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	No protocol
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	7
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	7
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	7
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	7
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	8
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	7-8
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	8
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	8
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	8

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Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	8
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	Not Applicable
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	9
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	9
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	9, and Appendix 4
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	10-13
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	Not Applicable
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	Not Applicable
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	15-16
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	17
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	17
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	17

From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(6): e1000097. doi:10.1371/journal.pmed1000097

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